

Why Earth's First Animals Were Stuck in Slow-Mo for Millions of Years

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Have you ever seen a strawberry plant send out runners to make new plants? That's exactly how some of Earth's first animals made babies for millions of years! And it turns out, that was a big reason why life stayed kind of boring for so long.

A new study from the University of Cambridge looked at weird, fern-like creatures called Fractofusus that lived over 500 million years ago. These animals didn't have mouths or guts. They just soaked up food from the water. And when they **reproduced**, they grew long runners (like strawberry plants) that sprouted identical copies of themselves.

"Life was pretty nice during the Ediacaran, so the need for sex was rather limited," said Dr. Emily Mitchell, who led the study. Because these clones were all connected, they shared **nutrients** and didn't have to compete. That sounds friendly, but it meant **evolution** was stuck in slow motion. No **competition**, no big changes.

The scientists used lasers and **artificial intelligence** to study fossils from Newfoundland, Canada. They even ran thousands of computer **simulations** to see how these ancient communities worked. The results showed that when animals finally started spreading out and having sex, diversity exploded. Stressful environments like storms and changing temperatures may have pushed them to try something new.

So, the next time you see a strawberry patch, remember: those runners once ruled the world, until sex changed everything!

Word Treasure

reproduced -- Made new copies of themselves, like a plant growing a new baby plant.

nutrients -- The food and minerals living things need to grow and stay healthy.

competition -- When living things struggle against each other for food, space, or mates.

evolution -- The slow change in living things over millions of years to become new forms.

artificial intelligence -- Computer programs that can learn and solve problems like a human brain.

simulations -- Computer-made copies of real events that help scientists test ideas.

Background you need

The University of Cambridge (a famous university in England) led this study of ancient life. These creatures lived during the Ediacaran period (a time long before dinosaurs, about 635-541 million years ago), when all life was in the ocean.

Fractofusus were among the first complex multicellular organisms on Earth, but they had no mouths, brains, or legs -- they simply absorbed food from the water around them.

Break it down

WHO: Scientists from the University of Cambridge, led by Dr. Emily Mitchell.

WHAT: They discovered that early animals like Fractofusus reproduced by cloning with runners, which slowed evolution for millions of years.

WHERE: Fossils were studied from Newfoundland, Canada, and computer simulations ran in a lab.

WHY: To understand why life stayed simple for so long and what finally triggered big changes.

Why it matters

Just like in a classroom or sports team, a little bit of challenge and mixing with different people helps you grow and get better -- life works the same way.

Test what you remember

1. What did Fractofusus use to make new copies of themselves?

- ☐ (A) Eggs and sperm
- ☐ (B) Long runners like strawberry plants
- ☐ (C) Seeds that floated in water
- ☐ (D) Splitting in half like bacteria

2. Why was evolution stuck in slow motion for early animals?

- ☐ (A) They had no food
- ☐ (B) They were too big to move
- ☐ (C) Clones didn't compete, so there was little change
- ☐ (D) The water was too cold

3. What did scientists use to study the fossils from Newfoundland?

- ☐ (A) Microscopes and hammers
- ☐ (B) Lasers and artificial intelligence
- ☐ (C) Diving suits and underwater cameras
- ☐ (D) Space telescopes

4. What happened when animals started spreading out and having sex?

- ☐ (A) They became weaker
- ☐ (B) They all died out
- ☐ (C) Diversity exploded
- ☐ (D) They stopped needing food

5. What does the article compare the first animals to?

- ☐ (A) Apple trees
- ☐ (B) Strawberry plants
- ☐ (C) Cactuses
- ☐ (D) Seaweed

6. Where were the fossils of Fractofusus found?

- ☐ (A) England
- ☐ (B) Australia
- ☐ (C) Newfoundland, Canada
- ☐ (D) Antarctica

Answer key (try the quiz first!): 1: B 2: C 3: B 4: C 5: B 6: C

Questions to think about

- 1. Cooperative View:** Cloning and sharing nutrients meant early animals lived in peaceful, connected colonies without fighting over food.
- 2. Diversity View:** Without competition and sexual reproduction, evolution slowed down, and life stayed boring and unchanging for millions of years.
- 3. Stress as Trigger:** Harsh environments like storms and temperature shifts may have finally pushed animals to try sexual reproduction, sparking the explosion of diversity we see today.

MY THOUGHTS (at least 20 words):